

COURSE: CS 355 — Reinforcement Learning

Credits: 3 | Department: Computer Science

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PREREQUISITES: CS 352

CO-REQUISITES: None

OFFERED: Fall (alternate years — offered in odd-numbered Fall semesters)

DESCRIPTION:

Foundations and algorithms of reinforcement learning. Topics include Markov Decision Processes (MDPs), dynamic programming (value iteration, policy iteration), Monte Carlo methods, temporal-difference learning (Q-learning, SARSA), function approximation, deep Q-networks (DQN), policy gradient methods (REINFORCE, PPO, A3C), and multi-agent RL intro. Labs use OpenAI Gym environments.

GRADING: Programming assignments 50%, Paper presentation 20%, Final project 30%.

NOTE: CS 355 is only offered in Fall semesters of odd-numbered years. Next offering: Fall 2025. Students should plan accordingly.